

AI Summary

■ Executive Overview This document is about the Rag Agentic AI Internship Training Program, specifically covering Module 16 and 17. The main objective is to understand retrieval, augmented generation, and semantic search applications using embeddings. The overall learning outcome is to build production-ready AI systems.

■ Major Topics * Retrieval Augmented Generation * Semantic Search Applications * Embeddings * Vector Databases * Agentic AI Workflows * Building AI Applications with FastAPI * Creating Intelligent Research Assistants

■ Key Concepts

Retrieval Augmented Generation Rag Rag is a powerful tool for retrieving relevant information from external documents. It uses embeddings to convert text into numerical vectors, allowing for efficient storage and retrieval of information.

Semantic Search Semantic search understands the meaning behind a query and matches it with relevant chunks in the database. This approach provides higher accuracy than traditional keyword-based searches.

Agentic AI Workflows Agentic AI agents think, plan, and use tools to complete tasks autonomously. They understand goals, plans tasks, make decisions, and remember information.

FastAPI and Building AI Applications FastAPI is a modern Python web framework used for building backend AI applications. It provides features such as automatic API documentation, dependency injection, and error handling.

■ Important Definitions

- * Embeddings: Convert text into numerical vectors for efficient storage and retrieval.
- * Vector Databases: Store embeddings instead of plain text for fast search queries.
- * Rag: A tool for retrieving relevant information from external documents using embeddings.

■ Key Takeaways

- * Understand retrieval augmented generation rag and its applications.
- * Learn about semantic search and its advantages over traditional keyword-based searches.
- * Familiarize yourself with agentic AI workflows and their benefits.
- * Use FastAPI to build production-ready AI applications.
- * Create intelligent research assistants using Rag and vector databases.